SRG Geometry

Local Response, Neighbor Exchange, and Compression

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Abstract

Moving points can change both the strength and the identity of their selected neighbors. We analyze these two effects on the next amplitudes of the existing Tower II model at a single fixed pass. A three-particle example gives an exact response while the selected neighbors remain unchanged. We derive the normalized-weight differential and a sufficient geometric margin preserving selection. A seven-particle example isolates one fifth/sixth-neighbor exchange, and a general jump formula retains the change in the selected enhanced-addition term. The existing signed compressor preserves differences on a common unclipped branch and identifies distinct nonzero limits exactly on common signed plateaus. A fixed rescaling of the seven incoming amplitudes makes the entire amplitude output constant despite a nonzero precompression jump. This is not equality of complete moving-particle states. January's accepted fixed-grid capture cells provide a separate comparison. All claims are one-pass, conditional, exact-real deductions from unchanged source expressions; no trajectories, new simulations or claims of physical validation are supplied.

1 What changes when the points move?

Imagine that each point receives values from its nearest neighbors. Moving the points slightly can change the strength of those influences even when the same neighbors remain selected. A further small movement can exchange one selected neighbor for another. If the two candidates carry different values, an arbitrarily small geometric change can then create a finite amplitude difference. A final compression may preserve that difference or map the two results to the same value.

We compare *incoming configurations at the same incremented pass number*. The configurations are not successive states produced by the model. Their incoming amplitudes, frequencies and other parameters are held fixed while geometry is compared, except in the one explicitly rescaled input family of Section 6.2. Geometry comparison is not physical time.

Model	Carrier and interaction	Meaning of equal output
January	A finite periodic field; fixed index neighbors; selected signed mode.	Equality of the complete field carrier.
Tower II	Moving particles with positions, amplitudes and velocities; directed distance-selected neighbors.	Equality of amplitudes alone leaves geometric state information separate.

These models are not composed into a replacement evolution. This paper presents the accepted OC03 and OC04 deductions in a single exposition [2, 3]. Their source is the frozen *Golden Frequency Tower—Part II* program [1]; the January comparison reuses accepted OC02 results [4]. The historical reconstruction in SRG-I and SRG Origins remains distinct [7, 8]. The separate proposed SRG-II mechanism paper is not renamed or replaced by this chapter.

2 Three points: response with the same neighbors

Place three particles on a line, with zero second and third coordinates:

Particle	Left	Middle	Right
Position	-1	s, $-1 < s < 1$	$+1$
Incoming amplitude	$1/10$	$1/5$	$3/10$

There is one frequency band, with frequency $\omega = 0.244$, no horizon and no breathing of the compression bound. All other parameters keep their defaults. The actual selector requests five neighbors; here it selects *both available other particles* in every row. It is not replaced by a two-neighbor rule. These are specified input data, not the output of an initializer [1, lines 34, 62–103, 111–127].

At $s = 0$ the middle particle receives equal influence from the two sides. At $s = 1/4$ the right particle is closer, so its larger amplitude receives more weight. With the source regularizer $\delta = 10^{-8}$, the two raw influences on the middle particle are

$$u_L = \frac{1}{1 + s + \delta}, \qquad u_R = \frac{1}{1 - s + \delta}.$$

Their normalized right share and the resulting average are

$$w_R = \frac{u_R}{u_L + u_R} = \frac{1 + s + \delta}{2(1 + \delta)} = \frac{1}{2} + \frac{s}{2(1 + \delta)}, \tag{2.1}$$

$$\overline{R}_M = (1 - w_R)\frac{1}{10} + w_R\frac{3}{10} = \frac{1}{5} + \frac{s}{10(1 + \delta)}. \tag{2.2}$$

Normalization changes the whole denominator; it cannot be held fixed while the distances change.

Every off-diagonal distance is positive and at most two. Thus each row's raw weight sum is at least $2/(2+\delta) > 10^{-10}$ and the amplitude update is active. Equal frequencies make the frequency factors one and fail the strict enhanced-mixing guard $|\omega_i - \omega_j| > 0.01$. The enhanced coupling remains at its default value; the guard makes its additions zero [1, lines 156–188].

For the moment collect the sine and later H multipliers into A_t:

$$H(t) = 2e^{-t/2}\cos t + e^{-3t/10}\sin t + \tfrac{3}{2}e^{-7t/10} + e^{-2t/5}, \tag{2.3}$$

$$a(t) = 1 + 0.01\sin(2\pi\omega t), \qquad b_t = 1 + 0.05H(t)/H_{\max}, \qquad A_t = a(t)b_t. \tag{2.4}$$

Here $t \geq 1$ is the incremented integer pass. The retained exact bound $|H(t)| \leq 9/2$ for $t \geq 0$, with any fixed finite $H_{\max} \geq 9/2$, gives

$$0.99 \leq a(t) \leq 1.01, \quad 0.95 \leq b_t \leq 1.05, \quad 0.9405 \leq A_t \leq 1.0605. \tag{2.5}$$

The source uses a sampled normalizer whose grid includes zero and then adds 10^{-12}. Its retained exact-real interpretation is $9/2 + 10^{-12}$. We have not executed the normalizer or freshly certified its binary arithmetic. Appendix B recalls the accepted bound without sampling.

Let q mean the amplitude immediately before compression. With $\beta = 0.15$, the source's own-amplitude/neighbor combination gives

$$q_M(s) = A_t\left[(1 - \beta)\frac{1}{5} + \beta\overline{R}_M(s)\right] = A_t\left[\frac{1}{5} + \frac{0.015s}{1 + \delta}\right]. \tag{2.6}$$

The full update order, and why positional stages leave this readout unchanged, are given in Section 3.

Lemma 2.1 (Uniform branch bound)**.** *In this equal-frequency example, every precompression amplitude belongs to* $[0.09405, 0.31815]$*, throughout* $-1 < s < 1$*. Every output lies on the positive unclipped branch of the source compressor.*

Proof. Each normalized received average is a convex combination of amplitudes between 0.1 and 0.3. Since $0 < \beta < 1$, combining it with the own amplitude preserves this interval. Multiplying by (2.5) gives

$$\frac{1881}{20000} = 0.09405 \le q_j \le 0.31815 = \frac{6363}{20000}. \tag{2.7}$$

The compressor has $L = 9$ and $\varepsilon_T = 10^{-9}$. Since $e > 2$,

$$e^{-9} < \frac{1}{512} < 0.09405, \qquad 0.31815 + \varepsilon_T < 512 < e^9.$$

Therefore its logarithmic clip is inactive and its output is $q_j + \varepsilon_T$. This includes epsilon in the branch check. □

Subtracting the two middle outputs cancels the same epsilon:

$$\boxed{R_M^{\mathrm{next}}(s) - R_M^{\mathrm{next}}(0) = \frac{0.015 A_t}{1+\delta} s.} \tag{2.8}$$

The coefficient is positive. This is a geometric response with no selected neighbor exchange, established on the whole interval.

s	$q_M(s)$	$R_M^{\mathrm{next}}(s) - R_M^{\mathrm{next}}(0)$
$-1/4$	0.204144213419495	−0.003900844802657
0	0.208045058222152	0
1/4	0.211945903024809	+0.003900844802657

These rounded illustrations are transcribed from the saved OC03 scalar results at the single common clock $t = 1$, using $H_{\max} = 9/2 + 10^{-12}$. They were evaluated there at 80 decimal digits, without interval arithmetic, and were not recomputed for this manuscript. The proof does not depend on the number of saved examples.

3 The source contract

For the local question, write the incoming positions as $x_i \in \mathbb{R}^3$ and amplitudes as $R_i \in \mathbb{R}$. Assigned frequencies $\omega_i > 0$, band identities, the tower-frequency list and parameters are fixed. Positions and velocities also belong to the particle state; the present readout is only its next amplitude vector.

Without a horizon and without collisions, the source sorts distances in each row, drops the zero self-entry and retains up to five other particles. Selection is directed, not symmetrized. With a horizon r_H, eligible particles satisfy the strict test $d_{ij} < r_H$. We write the selected set as $\mathcal{N}_i$ and $d_{ij} = \|x_j - x_i\|$.

The pass order matters [1, lines 129–215]:

1. Increment the pass number and, if enabled, set the breathing bound from incoming amplitudes. Breathing is false throughout the present analysis.
2. Store the incoming adjacency and distances, then move positions using the selected neighbors.

3. Couple amplitudes using those stored distances and incidences. Apply the sine multiplier, then the enhanced additive mixing.
4. Multiply amplitudes by the H factor.
5. Fuse positions within bands using the precompression amplitudes.
6. Compress amplitudes, then add positional noise and record velocities.

In particular, within-pass spatial movement does not refresh the weights used by that same amplitude calculation. Fusion changes positions only, and noise is added only to positions. Neither modifies the amplitude readout after its indicated stage.

For selected neighbors, the actual raw weights and normalization are

$$\chi_{ij} = \exp[-c(\log(\omega_j/\omega_i))^2], \qquad u_{ij} = \frac{\chi_{ij}}{d_{ij}+\delta}, \tag{3.1}$$

$$S_i = \sum_{j\in\mathcal{N}_i} u_{ij}, \qquad w_{ij} = \frac{u_{ij}}{S_i}, \qquad \overline{R}_i = \sum_{j\in\mathcal{N}_i} w_{ij} R_j. \tag{3.2}$$

The defaults are $c = 3$, $\delta = 10^{-8}$, $\beta = 0.15$, $L = 9$ and $\varepsilon_T = 10^{-9}$. A row with no neighbors or with $S_i < 10^{-10}$ is skipped at the mixing stage: its sine and enhanced additions are both skipped. Equality $S_i = 10^{-10}$ is active in the code, but provides no open active margin. Our local theorems require strict margins above that threshold [1, lines 151–163]; see also [5, A03].

For an active row define $a_i(t) = 1 + 0.01\sin(2\pi\omega_i t)$ and use b_t from (2.4). The source order gives

$$q_i = b_t \left\{ a_i(t)\big[(1-\beta)R_i + \beta\overline{R}_i\big] + \sum_{j\in\mathcal{N}_i} \eta_{ij} R_j \right\}. \tag{3.3}$$

The coefficients η_{ij} are determined by the source frequency tests, not fitted coefficients. Appendix A gives their full expression, including every strict guard and loop multiplicity. The sine does *not* multiply the later additive sum. For a skipped row, the corresponding precompression amplitude is instead $b_t R_i$; we do not extend (3.3) through such a boundary.

4 Smooth response inside a neighbor region

Let positions depend differentiably on a geometry parameter τ. A dot denotes differentiation with respect to τ, not time. Hold all incoming amplitudes, assigned frequencies, bands, tower list, clock and parameters fixed. Work on an open region with nonzero pair distances, fixed selected incidences and $S_i > 10^{-10}$.

Theorem 4.1 (Normalized geometry response, OC03)**.** *For any active finite selected row,*

$$\dot{\overline{R}}_i = -\sum_{j\in\mathcal{N}_i} w_{ij}(R_j - \overline{R}_i)\frac{\dot{d}_{ij}}{d_{ij}+\delta}, \tag{4.1}$$

$$\dot{d}_{ij} = \frac{x_j - x_i}{d_{ij}} \cdot (\dot{x}_j - \dot{x}_i), \tag{4.2}$$

$$\dot{q}_i = b_t a_i(t)\beta\,\dot{\overline{R}}_i. \tag{4.3}$$

These are statements about the conditional precompression map. The final amplitude derivative additionally requires a smooth compressor branch.

Proof. Put $\ell_{ij} = \dot{d}_{ij}/(d_{ij} + \delta)$. The fixed frequencies give $\dot{u}_{ij} = -u_{ij}\ell_{ij}$ and $\dot{S}_i = -S_i \sum_k w_{ik}\ell_{ik}$. Differentiating the shared denominator,

$$\dot{w}_{ij} = \frac{\dot{u}_{ij}S_i - u_{ij}\dot{S}_i}{S_i^2} = -w_{ij}\left(\ell_{ij} - \sum_k w_{ik}\ell_{ik}\right). \tag{4.4}$$

Multiplying by R_j and summing yields (4.1), since the amplitudes are fixed. Differentiating $d_{ij}^2 = (x_j - x_i)\cdot(x_j - x_i)$ gives (4.2). All η_{ij} and their selected incidences are fixed in this region. Differentiation of (3.3) therefore gives (4.3). □

If every selected neighbor has the same amplitude, every contrast $R_j - \overline{R}_i$ vanishes. Unequal amplitudes permit a response but do not guarantee one. For example, when all selected ℓ_{ij} equal one common value, the response vanishes because $\sum_j w_{ij}(R_j - \overline{R}_i) = 0$. Common translation is a special case with all distance changes zero.

Proposition 4.2 (A sufficient selection margin, OC03)**.** *Suppose a row has at least six other eligible candidates, fixed horizon membership and fifth/sixth distance gap $\Delta_i = d_{i,(6)} - d_{i,(5)} > 0$. If each particle displacement has norm at most ρ, then*

$$4\rho < \Delta_i \tag{4.5}$$

preserves its selected set, provided collisions and horizon/active-guard crossings are independently excluded.

Proof. For perturbations e_i, the reverse triangle inequality gives

$$|d'_{ij} - d_{ij}| \le \|e_j - e_i\| \le 2\rho.$$

For any originally selected j and excluded eligible k,

$$d'_{ik} - d'_{ij} \ge (d_{i,(6)} - 2\rho) - (d_{i,(5)} + 2\rho) = \Delta_i - 4\rho > 0.$$

Thus no excluded candidate can overtake a selected one. □

Useful independent sufficient margins are

$$2\rho < \min_{i\neq j} d_{ij} \qquad \text{(no collisions)}, \tag{4.6}$$

$$2\rho < \min_{i\neq j} |d_{ij} - r_H| \qquad \text{(fixed strict horizon membership)}, \tag{4.7}$$

$$\sum_{j\in\mathcal{N}_i} \frac{\chi_{ij}}{d_{ij} + \delta + 2\rho} > 10^{-10} \qquad \text{(active row)}. \tag{4.8}$$

The horizon condition is needed only when a horizon exists. The last expression is a lower bound on the perturbed raw weight sum once incidences are preserved. A geometric cutoff gap alone does not ensure activity. The condition $4\rho < \Delta_i$ is sufficient, not necessary or claimed sharp. With at most five eligible other particles, all are selected; no sixth distance exists and no cutoff gap is needed. Ties within a selected set are harmless to its incidence set, unlike a tie across its cutoff.

5 Seven points: an isolated neighbor exchange

Keep a focal particle at zero and four common neighbors to its right at $1/2, 1, 3/2, 2$. Two farther particles compete for the last selected place. The right candidate A, near $+3$, has amplitude 0.1; the left candidate B, near -3, has amplitude 0.3. Move both candidates slightly left: A is closer to the focal point. Move both slightly right: B is closer. Equal distances at the boundary do not make their incoming amplitudes equal.

Particle	B	i	c1	c2	c3	c4	A
Position	$-3+s$	0	1/2	1	3/2	2	$3+s$
Amplitude	3/10	1/5	1/5	1/5	1/5	1/5	1/10

All second and third coordinates are zero, $-1/4 < s < 1/4$, and the seven particles have one band at frequency 0.244, no horizon, no breathing, and unchanged default parameters. This is a specified incoming family.

Proposition 5.1 (Isolation in the full directed graph, OC04)**.** *For $s < 0$ the focal row selects A and excludes B; for $s > 0$ it selects B and excludes A. At zero these candidates tie for fifth/sixth place. Every other row retains its selected set throughout the interval. Every row has a strict active-weight margin.*

Proof. The focal common distances are $1/2, 1, 3/2, 2$. The candidate distances $3+s$ and $3-s$ are both greater than $11/4 > 2$; their difference is $2s$, with slope 2. They exchange order transversely without a collision.

At any common coordinate $x \in \{1/2, 1, 3/2, 2\}$, B is farther than every focal/common particle, since $x + 3 - s > 13/4$ while those distances are at most 2. B is also farther than A, because

$$(x + 3 - s) - (3 + s - x) = 2(x - s) > 0.$$

Hence every common row uniquely excludes B. The A–B distance is 6. All other distances from A are at most $3 + s < 13/4$, so A excludes B; all other distances from B are at most $5 - s < 21/4$, so B excludes A. All positions are distinct and all pair distances are at most 6. Each row selects five equal-frequency inverse-distance weights, giving

$$S_i \geq \frac{5}{6+\delta} = \frac{500000000}{600000001} > 10^{-10}. \tag{5.1}$$

This proves isolation of the focal exchange and strict activity on the whole interval. □

At the switch define

$$S = \sum_{k=1}^{4} \frac{1}{k/2+\delta}, \qquad u_* = \frac{1}{3+\delta}, \qquad w_* = \frac{u_*}{S+u_*}. \tag{5.2}$$

The last quantity is the normalized share of the fifth neighbor. Equal frequencies again turn off the guarded enhanced additions without changing their parameters. The limiting received averages are $1/5 - w_*/10$ and $1/5 + w_*/10$. With A_t from (2.4),

$$q_- = A_t\left(\frac{1}{5} - \frac{3}{200}w_*\right), \qquad q_+ = A_t\left(\frac{1}{5} + \frac{3}{200}w_*\right), \tag{5.3}$$

$$\Delta q_i = q_+ - q_- = \frac{3}{100}A_t w_* > 0. \tag{5.4}$$

All seven incoming amplitudes lie in $[0.1, 0.3]$, so the convex argument of Lemma 2.1 applies unchanged. All outputs are positive and unclipped. The same epsilon cancels and the compressed jump equals (5.4).

The regularized share is approximately 0.0740740747828074933. Only the formal $\delta \to 0$ comparison gives $S = 25/6$, $u_* = 1/3$ and $w_* = 2/27$. We do not substitute that comparison for the actual source weight. The retained $t = 1$ scalar results give

$$q_- \approx 0.206889252332082, \qquad q_+ \approx 0.209200864112223, \qquad \Delta q_i \approx 0.002311611780141188.$$

The source specifies a sorting call, not a separately defined portable exact-tie rule. No tie-sorting experiment is needed or supplied. The two one-sided limits are independent of the tie choice. Because they differ, no choice of the single tie value can make this amplitude map continuous there.

5.1 The general exchange law

Let the focal selected sets on two sides of a boundary be $C \cup \{A\}$ and $C \cup \{B\}$, where C contains four unchanged neighbors. Assume $d_{iA} = d_{iB} = d_* > 0$ at the boundary and common limits for other distances. Keep the frequencies, incoming amplitudes, bands, tower list, parameters and clock fixed. Assume a transverse candidate-order exchange, no further simultaneous focal cutoff, horizon or guard event, and both limiting raw weight sums strictly above 10^{-10}. These margins allow (3.3) on both sides.

At the boundary put $u_j = \chi_{ij}/(d_{ij} + \delta)$, $S = \sum_{j \in C} u_j > 0$ and $V = \sum_{j \in C} u_j R_j$. Normalize separately:

$$\overline{R}_- = \frac{V + u_A R_A}{S + u_A}, \qquad \overline{R}_+ = \frac{V + u_B R_B}{S + u_B}. \tag{5.5}$$

Theorem 5.2 (Active-row exchange, OC04)**.** *Under these assumptions the one-sided differences satisfy*

$$\Delta \overline{R}_i = \frac{u_B}{S + u_B}\left(R_B - \frac{V}{S}\right) - \frac{u_A}{S + u_A}\left(R_A - \frac{V}{S}\right), \tag{5.6}$$

$$\Delta q_i = b_t \left[a_i(t)\beta\, \Delta \overline{R}_i + \eta_{iB} R_B - \eta_{iA} R_A\right]. \tag{5.7}$$

Since $b_t > 0$ under the retained defaults, the exact zero-jump condition is

$$a_i(t)\beta \left[\frac{V + u_B R_B}{S + u_B} - \frac{V + u_A R_A}{S + u_A}\right] + \eta_{iB} R_B - \eta_{iA} R_A = 0. \tag{5.8}$$

Proof. Subtract V/S from each fraction in (5.5); this gives (5.6). Subtract the two source-ordered expressions (3.3). The own-amplitude baseline and every common-neighbor additive term cancel, leaving (5.7). The lower bound $b_t \geq 0.95$ then gives (5.8). □

An additive coefficient is independent of distance but its *selected identity* changes here. Its zero positional derivative inside a fixed neighbor region does not justify discarding the selected term at a switch. This is the boundary distinction between Theorems 4.1 and 5.2.

Equal candidate distances need not give $u_A = u_B$, because the frequency matches may differ. Thus the two denominators generally differ. If $R_A = R_B = r$, the received-average contrast is

$$\left(\frac{u_B}{S + u_B} - \frac{u_A}{S + u_A}\right)\left(r - \frac{V}{S}\right),$$

which need not vanish. Even when all selected incoming amplitudes equal $r > 0$, the remaining jump can be $b_t r(\eta_{iB} - \eta_{iA})$. A same-frequency A has coefficient zero, while an available adjacent φ-band B has a strictly positive coefficient under the source guard.

Conversely, unequal amplitudes do not guarantee a jump. For example, when $V = 0$, define $K_A = a_i \beta u_A/(S + u_A) + \eta_{iA} > 0$ and similarly $K_B > 0$. Whenever $K_A \neq K_B$, choosing $R_B = K_A R_A / K_B$ with $R_A > 0$ makes (5.8) zero despite unequal candidates. This is a conditional algebraic illustration, not an extra numerical fixture. In the equal-frequency example, $\eta = 0$ and $u_A = u_B$, so the law reduces to $\Delta q_i = A_t \beta w_* (R_B - R_A)$ and gives (5.4).

6 What the existing compressor identifies

The frozen source applies

$$C_T(q) = \operatorname{sgn}(q) \exp\left[\operatorname{clip}\left(\log(|q| + \varepsilon_T), -9, 9\right)\right], \qquad \varepsilon_T = 10^{-9}. \tag{6.1}$$

Set $u_T = e^{-9}$, $v_T = e^{9}$, $f_T = u_T - \varepsilon_T > 0$ and $h_T = v_T - \varepsilon_T$. Monotonicity of logarithm and exponential gives the complete exact-real branches:

$$C_T(q) = \begin{cases} 0, & q = 0, \\ \operatorname{sgn}(q) u_T, & 0 < |q| \le f_T, \\ q + \varepsilon_T \operatorname{sgn}(q), & f_T < |q| < h_T, \\ \operatorname{sgn}(q) v_T, & |q| \ge h_T. \end{cases} \tag{6.2}$$

The floor upper edge and ceiling lower edge are included; zero is a separate case [1, lines 42–44, 208–209].

Theorem 6.1 (Nonzero limiting inputs, OC04)**.** *For finite nonzero one-sided precompression limits q_-, q_+, the compressed jump is $C_T(q_+) - C_T(q_-)$. A common signed unclipped branch preserves the input difference exactly. Distinct such limits have equal outputs if and only if both lie on one common signed floor or ceiling plateau. Opposite nonzero signs cannot have the same output.*

Proof. The branches are continuous at every nonzero input. At the positive floor edge the middle expression gives $f_T + \varepsilon_T = u_T$; at the positive ceiling edge it gives $h_T + \varepsilon_T = v_T$. Odd reflection gives the negative edges. Hence continuity passes each nonzero limit through C_T. The clipping edges are nondifferentiable, not discontinuous.

On either fixed signed unclipped branch the same signed epsilon cancels under subtraction. On the positive side, the fibers of u_T and v_T are exactly $(0, f_T]$ and $[h_T, \infty)$; between them the function is strictly increasing and has singleton fibers. The negative plateau fibers are $[-f_T, 0)$ and $(-\infty, -h_T]$, with singleton fibers in between. The output signs separate the two sides. These exhaust the possibilities. □

For smooth incoming comparisons, the final amplitude derivative is $\dot{q}_i$ on a nonzero unclipped branch and zero on the interior of a fixed signed plateau. This statement does not extend a derivative through the clipping edges or zero.

6.1 Zero requires an approach condition

If precompression values tend to zero and are eventually strictly positive, they eventually belong to $(0, f_T]$, so their compressed limit is $+u_T$. Eventually strictly negative approach gives $-u_T$. Values identically zero on a side near the boundary remain zero. Thus $C_T(0) = 0$ cannot be substituted for an unqualified limit. If none of these approach conditions is established, a zero precompression limit alone does not establish a compressed limit. We make no general oscillatory-boundary classification.

6.2 A whole amplitude vector on one floor

The unscaled seven-point family has the positive jump (5.4), preserved by its unclipped branch. Now keep the same positions, selected rule, frequency, clock and model parameters but multiply every incoming amplitude by the single fixed data scale $\kappa = 10^{-4}$. This is a second specified input family, not a parameter sweep or a modified operator.

The equal-frequency additions still vanish and the remaining amplitude expression is linear in incoming amplitudes. Because L does not breathe, precompression amplitudes scale by κ without changing the compressor. The uniform bound (2.7) therefore gives

$$0 < q_j^{(\kappa)}(s) \le \kappa \frac{6363}{20000}, \qquad \kappa \frac{6363}{20000} + 10^{-9} = \frac{3977}{125000000} < \frac{1}{19683} < e^{-9}. \tag{6.3}$$

The first strict difference is the positive rational number 46720709/2460375000000; the last inequality follows from $e < 3$. For completeness, $e = \sum_{n\geq 0} 1/n! > 2$, and $n! \geq 2^{n-1}$ for $n \geq 2$, strictly for $n \geq 3$, gives $e < 2 + \sum_{n\geq 2} 2^{-(n-1)} = 3$.

Every row is therefore strictly on the positive floor throughout the whole interval, including either focal choice at the tie:

$$\boxed{(R_1^{\mathrm{next}}, \ldots, R_7^{\mathrm{next}}) = (e^{-9}, \ldots, e^{-9}).} \tag{6.4}$$

The focal precompression jump remains the strictly positive $\kappa(3/100)A_t w_*$, but the compressed jump is zero. The saved $t = 1$ values are $q_-^{(\kappa)} \approx 0.0000206889252332082$ and $q_+^{(\kappa)} \approx 0.0000209200864112223$; their difference is approximately $2.311611780141188 \times 10^{-7}$ before compression and zero afterward. These retained values illustrate the uniform proof; they are not a seven-particle update.

Equation (6.4) is *not full particle-state capture*. Band fusion has already used the precompression amplitudes through both its weighted center and its positional movement factor. Positions and velocities carry separate information. No equality or inequality of complete final particle states, and no later trajectory consequence, follows automatically from this amplitude theorem.

7 January: a separate fixed-grid reference

January's accepted signed-mode analysis has a finite periodic field $X \in \mathbb{R}^{NM}$ as its carrier. This signed mode is selected for the analysis; the source app initially selects paper mode. Its fixed index stencil is

$$\begin{aligned}(\mathcal{L}X)_{ij} &= X_{i+1,j} - 2X_{ij} + X_{i-1,j} \\ &\quad + \frac{25}{9}(X_{i,j+1} - 2X_{ij} + X_{i,j-1}),\end{aligned} \tag{7.1}$$

$$T_G = I + \frac{1}{20}\mathcal{L}, \qquad P(X) = T_G X + \frac{1}{10}\tanh X, \qquad F_m(X) = C_s(mP(X)). \tag{7.2}$$

Here T_G denotes the operator called T in OC02; the subscript avoids confusion with the Tower II source key. The tanh and compressor act sitewise. Duplicate incidences for axes of length two are counted twice; on an axis of length one the corresponding neighbor is the site itself. This is the original periodic index stencil, not an embedded-surface Laplacian. The constants are $L = 10$ and $\varepsilon_G = 10^{-6}$:

$$C_s(q) = \operatorname{sgn}(q)\,\operatorname{clip}(|q| + \varepsilon_G, e^{-10}, e^{10}) \quad (q \neq 0), \qquad C_s(0) = 0.$$

These definitions are retained from OC02 [4, equations (1)–(2)] and its source correspondence [6]. For a sign field $\sigma_{ij} \in \{-1, 1\}$ and a fixed gain $m > 0$, the accepted one-step capture condition is

$$F_m(X) = e^{-10}\sigma \quad \Longleftrightarrow \quad 0 < \sigma_{ij} m P(X)_{ij} \leq e^{-10} - 10^{-6} \quad \text{for every site.} \tag{7.3}$$

Indeed, the signed floor has exactly that nonzero half-open scalar preimage at each site. The zero face is excluded; the upper face is included.

OC02's retained Theorem 2 states that P is a global smooth diffeomorphism. Its mechanism is that the symmetric stencil has lower quadratic-form bound 11/45, so $DP = T_G + \operatorname{diag}(\operatorname{sech}^2 X)/10$ is uniformly positive definite. The strictly convex, coercive potential $X^T T_G X/2 + \sum_{ij} \log\cosh(X_{ij})/10$ supplies a unique solution of $P(X) = z$ for every z, and the inverse function theorem supplies the smooth inverse. This recalls the accepted argument, not a new classification.

Writing $b = (e^{-10} - 10^{-6})/m$, the capture set is consequently

$$P^{-1}\{z : 0 < \sigma_{ij} z_{ij} \leq b \text{ at every site}\},$$

the inverse image of a half-open rectangular cell. The interior uses strict inequalities on both faces; the closure includes zero. A field may reach an arbitrary signed floor target in one step without that target persisting. Persistence under subsequent gains requires the accepted common-fixed-pattern conditions. We do not recount that classification or compute multistep preimages [4, lines 224–269].

Both models can erase a distinction at compression. Equality of January's complete field carrier and equality of Tower II's amplitude vector are, however, different mathematical statements. The latter leaves moving geometry outside the equality. No January basin or fixed-pattern theorem has been transferred to particle states.

8 Scope, evidence and interpretation

The results concern exact-real algebra of the unchanged source expressions at one fixed pass. Smooth response is conditional on an active region with unchanged incidences and nonzero distances; the exchange theorem concerns one isolated focal event with strict limiting margins. Final readout statements retain the compressor's sign and branch conditions. Not every exchange jumps, and compression does not erase every difference.

The H bound and January results are inherited mathematics. The geometry response, selection margin, exchange law and constructive compressor examples are OC03/OC04 deductions in this research program. Their proofs are reproduced here. No literature survey has established global novelty or priority over general results on normalized weights, nearest-neighbor maps or clipping. No physical validation is claimed.

OC03 and OC04 each retain eight mathematical or scalar groups and two groups for input identity and preservation. Their saved exact rational and symbolic identities and 80-digit, non-interval scalar corroborations support examples; they do not replace the proofs. Neither checker was rerun for this paper. GPT's separate OC04 review reports independent symbolic and 100-digit scalar corroboration, without executing either checker or historical program. That review is not external peer review, a binary execution certificate or a replacement for the saved evidence.

No source normalizer, initializer, particle update, trajectory or stochastic step was executed in preparing this edition. New QA concerns transcription, source mapping, typography and the review artifacts only. There are no arrival statistics, autonomous-attractor results, general recursion theorem or full-state capture theorem here. Later-pass questions remain future work; they do not weaken the stated one-pass conclusions.

Conclusion

Incoming geometry affects the next amplitude in two distinct ways: smooth variation of normalized influences within a selected-neighbor region, and a possible finite jump when the selected identity changes. The jump law keeps both the normalized average and the switched enhanced addition. The existing compressor then preserves a common signed unclipped difference or identifies a common signed plateau. The two explicit geometric examples make these alternatives checkable. They establish a local amplitude account while keeping the complete moving-particle state, later dynamics and January's separate field carrier in their proper scopes.

Assistance, status and reuse

Hilmir Frímann Halldórsson is the named human author. GPT supplied mathematical review of OC03/OC04 and the manuscript work order. Codex performed the recorded OC03/OC04 analysis and finite checks, then assembled this manuscript, checked its source correspondence and completed artifact QA. Those assistance roles are not coauthorship or external peer review. The underlying analyses are internally accepted; this new v0.1 manuscript and its proposed public payload await GPT review. No institutional affiliation, DOI or series number is assigned.

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A The complete enhanced-addition coefficient

Let $\nu = 0.001$, $\varphi = (1+\sqrt{5})/2$, $D_{ij} = |\omega_i - \omega_j|$, $U_{ij} = \omega_i + \omega_j$, and let $\mathcal{W}$ be the fixed source tower-frequency list. The coefficient in (3.3) is exactly

$$\eta_{ij} = \nu \mathbf{1}_{\{D_{ij}>0.01\}} \Bigg[\sum_{p\in\{-2,-1,1,2\}} e^{-5[\log(\omega_i/\omega_j) - p\log\varphi]^2} + \frac{1}{2} \sum_{\Omega\in\mathcal{W}} \left(\mathbf{1}_{\{|U_{ij}-\Omega|/\Omega<0.2\}} + \mathbf{1}_{\{D_{ij}>0.01\}} \mathbf{1}_{\{|D_{ij}-\Omega|/\Omega<0.2\}} \right) \Bigg]. \tag{A.1}$$

This transcribes lines 169–188 of the frozen source [1]. Each ratio test and each matching tower entry contributes separately. The repeated difference guard records the inner test; it is redundant once the outer guard holds. The whole coefficient is added after sine multiplication, as in (3.3). For identical frequencies the outer guard fails and $\eta_{ij} = 0$. Within fixed incidences the coefficient has no positional derivative, but a changed selected identity contributes the last two terms of (5.7).

B The retained H bound

For the same expression (2.3), the accepted argument [4, lines 185–195] uses $|\sin t| \le t$ to bound

$$|H(t)| \le B(t) + \tfrac{3}{2}e^{-7t/10} + e^{-2t/5}, \qquad B(t) = 2e^{-t/2} + te^{-3t/10}.$$

Its derivative is $B'(t) = -e^{-t/2} + (1 - 3t/10)e^{-3t/10}$. For $t \ge 10/3$ both summands are nonpositive. For $0 \le t < 10/3$, $1 - 3t/10 \le 1 - t/5 \le e^{-t/5}$, so again $B'(t) \le 0$. Thus $B(t) \le 2$ and $|H(t)| \le 2 + 3/2 + 1 = 9/2$, attained at zero. The source's 50,000-point grid on $[0, 100]$ includes zero; its exact-real maximum is therefore $9/2$, and its added guard gives $H_{\max} = 9/2 + 10^{-12}$. This inherited mathematical argument makes no claim of a fresh floating-point audit of that helper.

References

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